Scalability & Future Plan

Date	15 March 2024
Team ID	SWTID-2026-1529
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Rate Limiting on Free Tier Gemini API**	High concurrent traffic spikes may encounter temporary request throttling.	High
2	**Client-Side Local Storage for Study History**	Study notes and quiz logs are stored on browser storage rather than synced across multiple devices.	Medium
3	**Single-Region Backend Server Deployment**	Slight latency variation for international learners across diverse geographic regions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Uvicorn worker instance handling ~100 concurrent users.	Deploy horizontal auto-scaling container clusters on Kubernetes / Vercel Serverless Edge.
2	Data Storage	Supabase PostgreSQL cloud with browser-level storage.	Implement Redis caching layer for common academic prompts and full user profile synchronization in Supabase.
3	Performance	Synchronous JSON response cycle (~1.1s latency).	Introduce Server-Sent Events (SSE) streaming for real-time word-by-word AI tutor responses.
4	Security	Environment variable key protection and CORS filtering.	Enforce Supabase Row-Level Security (RLS), OAuth2 authentication, and API rate-limiting gateway.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Collaborative Study Rooms & Peer Quiz Battles with Live Leaderboards	Q3 2026	Drives peer-to-peer engagement and gamified classroom adoption.
Phase 3	Multilingual Educational Content Generation (Hindi, Telugu, Tamil, Spanish)	Q4 2026	Expands educational access to regional vernacular students globally.
Phase 4	Enterprise LMS Integrations (Canvas, Blackboard, Moodle LTI 1.3)	Q1 2027	Automated quiz syncing, assignment generation, and institution-wide grading.